EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Need of Classification of elements; Genesis of periodic classification; Lavoisier classification; Dobereiner's Triads; Newland's Law of Octaves; Lothar Meyer Curve

1. Which of the following set of elements obey Newland's octave rule

(A) Na, K, Rb (B) F, Cl, Br (C) Be, Mg, Ca (D) B, Al, Ga

CPT079

2. Which of the following is/are Dobereiner triad :-

(a) P, As, Sb (b) Cu, Ag, Au (c) Fe, Co, Ni (d) S, Se, Te

Correct answer is :-

(A) a and b (B) b and c (C) a and d (D) All

CPT080

3. The element that was present in Ascending position in Lothar – Meyer curve is .....

(A) Cl (B) Ca (C) Si (D) Mn

CPT085

4. Atomic weight of an element 'X' is 32 and that of element Z is 127. At wt. of their intermediate element Y, as per Dobereiner triad will be :

(A) 171 (B) 32 (C) 79.5 (D) 85.5

CPT086

5. The element which occupies peak of Lothar Meyer Curve is :

(A) V (B) Se (C) K (D) La

CPT087

Rankers Academy JEE

6. Select the incorrect statement for Lothar Meyer's curve :

(A) Curve is plotted between atomic weight and atomic volume

(B) Alkali metals occupy maxima of curve

(C) Halogens occupy descending portions of the curve

(D) Transition metals occupy bottom portions of the curve

CPT089

Mendeleev Periodic Table

7. Which is not anomalous pair of elements in the Mendeleev's periodic table:-

(A) Ar and K (B) Co and Ni (C) Te and I (D) Al and Si

CPT081

8. Which are correct match :-

(a) Eka silicon –Be (b) Eka aluminium – Ga

(c) Eka manganese – Tc (d) Eka scandium – B

(A) b, c (B) a, b, d (C) a, d (D) All

CPT082

Long Form of Periodic Table

9. The electronic configuration of elements X and Z are $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$ and $1s^2\ 2s^2\ 2p^5$ respectively. What is the position of element X with respect to position of Z in the periodic table -

(A) Just below element Z (B) Just above Z

(C) Left to the Z (D) right to the Z

CPT088

For More Join: @IITJEE_Advanced

10. If each orbital can hold a maximum of three electrons, the number of elements in 9th period of periodic table (long form) will be

(A) 48 (B) 162 (C) 50 (D) 75

CPT090

Nomenclature of Elements with Atomic Number >100; Notation for IUPAC Nomenclature of Elements

11. The IUPAC name of the element which is placed just after $_{105}Db$ is the periodic table, will be :-

(A) Unnilpentium (B) Ununnilium (C) Unnilhexium (D) Unnilquadium

CPT095

Electronic Configuration of Elements and the periodic table, Electronic Configuration in Period; Groupwise Electronic Configuration, Identification of group, block and period of an element

12. The electronic configuration of an element is $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^4$. The atomic number and the group number of the element 'X' which is just below the above element in the periodic table are respectively.

(A) 24 & 6 (B) 24 & 15 (C) 34 & 16 (D) 34 & 8

CPT091

13. The number of d- electrons in $Mn^{2\oplus}$ is equal to that of

(A) p-electrons in N (B) s-electrons in Na

(C) d-electrons in $Fe^{2\oplus}$ (D) p-electrons in $O^{2\ominus}$

CPT092

14. Which of the following has the maximum number of unpaired electrons

(A) $Mg^{2\oplus}$ (B) $Ti^{3\oplus}$ (C) $V^{3\oplus}$ (D) $Fe^{2\oplus}$

CPT093

Rankers Academy JEE

15. Lanthanoids belong to :

(A) V^{th} period (B) VIIth period (C) IIIrd group (D) IVth group

CPT097

Definition of Effective Nuclear Charge (Z_{eff}); Shielding Effect; Penetration Power

16. Find correct order of penetration power

(A) ns > nd > np > nf (B) nf > nd > np > ns

(C) ns > np > nd > nf (D) nf > np > nd > ns

CPT100

17. Correct order of Z_{eff} of different ions of an element (M) is:-

(A) $M^{\oplus} < M < M^{\ominus}$ (B) $M^{\oplus} > M > M^{\ominus}$ (C) $M^{\oplus} < M > M^{\ominus}$ (D) $M^{\ominus} > M^{\oplus} > M$

CPT103

18. Correct order of Z_{eff} is in following:-

(A) $Mg^{2\oplus} > Na^{\oplus} > F^{\ominus} > O^{2\ominus} > N^{3\ominus}$ (B) $Mg^{2\oplus} > Na^{\oplus} > N^{3\ominus} > O^{2\ominus} > N^{3\ominus}$

(C) $Mg^{2\oplus} < Na^{\oplus} < F^{\ominus} > O^{2\ominus} > N^{3\ominus}$ (D) $Mg^{2\oplus} < Na^{\oplus} < F^{\ominus} < O^{2\ominus} < N^{3\ominus}$

CPT104

Calculation of Shielding Constant using Slater's Rule

19. Value of Z_{eff} on Last $e^{\ominus}$ of N

(A) 3.5 (B) 3.8 (C) 3.9 (D) 3.25

CPT102

20. What is the value of effective nuclear charge for last electron of yttrium (Y) ?

(A) 36 (B) 3 (C) 39 (D) 9

CPT106

For More Join: @IITJEE_Advanced

21. Find the shielding constant (σ) by the first electron enter in $H^{\ominus}$?

(A) 0.70 (B) 1 (C) 0.35 (D) 0.30

CPT107

Atomic Radius: Definition; Types

22. For the element X, student Mansi Measured its radius as 102 nm, student Rohit as 203 nm and Ankur as 100 nm. Using same-apparatus. Their teacher explained that measurement were correct by saying that recorded value by three students were :-

(A) crystal, Van der Waals and covalent radii (B) Covalent, Crystal and Van der Waals radii

(C) Van der Waals, ionic and covalent radii (D) No is correct.

CPT112

Factors Affecting Atomic Radius; Periodic Trends for only s and p Block Elements

23. Arrange the following in order of increasing atomic radii Na, Si, Al, Ar :-

(A) Na < Si < Al < Ar (B) Si < Al < Na < Ar (C) Ar < Al < Si < Na (D) Na < Al < Si < Ar

CPT110

Ionic Radius; Factors Affecting Ionic Radius

24. The correct order of atomic or ionic size

(A) N < Li < B (B) Cl < Mg < Ca (C) $Ca^{2\oplus} < S^{2\ominus} < Cl^{\ominus}$ (D) $Na^{\oplus} < Mg^{2\oplus} < Cl^{\ominus}$

CPT179

Atomic Radius ; Trends for d- and f- Block Elements

25. The correct order of size would be:-

(A) Ni < Pd $\approx$ Pt (B) Pd < Pt < Ni (C) Pt > Ni > Pd (D) Pd > Pt > Ni

CPT113

Rankers Academy JEE

Definition of Ionization Energy; Successive Ionization Energy

26. (a) $M^{\ominus}_{(g)} \rightarrow M_{(g)}$ (b) $M_{(g)} \rightarrow M^{\oplus}_{(g)}$ (c) $M^{\oplus}_{(g)} \rightarrow M^{2\oplus}_{(g)}$ (d) $M^{2\oplus}_{(g)} \rightarrow M^{3\oplus}_{(g)}$

Minimum and maximum Ionisation Potential would be of:-

(A) a, d (B) b, c (C) c, d (D) d, a

CPT117

Factors Affecting Ionization Energy

27. Factor that has no effect on the ionisation potential:

(A) atomic size (B) atomic mass

(C) Effective nuclear charge (D) Shielding effect

CPT116

28. False statement for periodic classification of elements is

(A) The properties of the elements are periodic function of their atomic numbers.

(B) No. of non-metallic elements is less than the no. of metallic elements.

(C) First ionization energy of elements does not increase regularly with the increase in atomic number in a period.

(D) d-subshell is filled by final electron with increasing atomic number of inner transition elements.

CPT183

Periodicity of Ionization Energy in s and p Block Elements

29. Amongst the following, the incorrect statement is

(A) IE_1 (Al) < IE_1 (Mg) (B) IE_1 (Na) < IE_1(Mg) (C) IE_2 (Mg) > IE_2 (Na) (D) IE_3 (Mg) > IE_3 (Al)

CPT180

For More Join: @IITJEE_Advanced

Periodicity of Ionization Energy in d Block Elements

30. Correct order of I.P is :

(A) Zn > Mn (B) Xe > Ne (C) Sc > Mn (D) Na > As

CPT127

Applications of Ionization Energy; Electropositive Character; Metallic and Non Metallic Nature; Reducing Power and reactivity of metals

31. The strongest reducing agent among the following is :-

(A) Na (B) Mg (C) Al (D) K

CPT128

32. Which of the following element has highest metallic character.

Element	IP
(A) P	17 eV
(B) Q	2 eV
(C) R	10 eV
(D) S	13 Ev

CPT131

Definition of Electron Affinity and Electron Gain Enthalpy

33. If for an atom both ionization energy and electron affinity are low, atom will tend to ____electron rather than _____ electron ; and hence can be classified as _____element.

Select option to fill blank spaces respectively.

(A) gain ; lose ; electronegative (B) lose ; gain ; electropositive

(C) lose ; gain ; electronegative (D) gain ; lose ; electropositive

CPT133

Pankers Academy JEE

34. Electron affinity is a :-

(A) Relative strength to attract the shared electron pair

(B) Necessary energy required to remove the electron from the ultimate orbit

(C) Energy released when an electron is added to the outermost shell

(D) Energy released when an electron is added to the inner shell

CPT136

Periodicity of Electron Affinity in s and p block elements

35. Select the correct code in terms of true (T) and false (F), respectively.

(i) Process of adding electron to an atom may be exothermic or endothermic.

(ii) In most of the elements energy is released when an electron is added to the atom.

(iii) Halogens have maximum ΔH_{eg} in their respective period because of their stable electronic configuration.

(iv) Electron affinity order S < O

(v) ΔH_{eg} is maximum positive for inert gases in their respective period

(A) T F F T F (B) F T T F T (C) T T F F T (D) F F F T F

CPT137

36. Correct electron gain enthalpy order is :

(A) N < O < Cl < Al (B) O < N < Al < Cl (C) N < Al < O < Cl (D) Cl < N < O < Al

CPT139

37. Electron affinities of O, F, S and Cl are in the order.

(A) O < S < Cl < F (B) O < S < F < Cl (C) S < O < Cl < F (D) S < O < F < Cl

CPT140

For More Join: @IITJEE_Advanced

38. Which of the following statements is not true?
(A) F atom can hold additional electron more tightly than Cl atom
(B) Cl atom can hold additional electron more tightly than F atom
(C) The incoming electron encounters greater repulsion for F atom than for Cl atom
(D) It is easier to remove an electron from $F^{\ominus}$ than $Cl^{\ominus}$.

CPT182

2nd Electron Affinity

39. $O_{(g)} + 2e^{\ominus} \rightarrow O_{(g)}^{2\ominus}$; ΔH_{eg} = 744.7 KJ/mole. The positive value of ΔH_{eg} is due to :-
(A) Energy is released to add 1 $e^{\ominus}$ to $O^{1\ominus}$
(B) Energy is required to add 1 $e^{\ominus}$ to $O^{1\ominus}$
(C) Energy is needed to add 1$e^{\ominus}$ to O 1$e^{\ominus}$ to O
(D) None of the above is correct

CPT135

Relation between Ionization Energy and Electron Affinity

40. Ionisation energy of $F^{\ominus}$ is 320 kJ mol^{-1}. The electron gain enthalpy of fluorine would be :
(A) –320 KJ/mol (B) –160 KJ/mol (C) +320 KJ/mol (D) +160 KJ/mol

CPT138

Factors Affecting Electronegativity; Periodicity of Electronegativity in s-p- block elements

41. The correct set of decreasing order of electronegativity is :-
(A) Li, H, Na (B) Na, H, Li (C) H, Li, Na (D) Li, Na, H

CPT145

42. On the Pauling's electronegativity scale, which element is next to F.
(A) Cl (B) O (C) Br (D) Ne

Rankers Academy JEE

CPT153

Applications of Electronegativity; Polarity of Bond; Calculation of Percentage Ionic Character

43. % Ionic character of compound AB is
EN_A = 1.8 EN_B = 2.6
(A) 42.42% (B) 24.24% (C) 15.04% (D) 17.08%

CPT148

Applications of Electronegativity; Acidic Strength of Oxides and Hydroxides

44. Correct Order of acidic Character?
(A) ClOH < IOH < BrOH (B) ClOH < BrOH < IOH
(C) IOH < BrOH < ClOH (D) None of these

CPT161

45. In periodic table, the basic character of oxides ?
(A) Increase from left to right and decreases from top to bottom
(B) decreases from right to left and increases from top to bottom
(C) decreases from left to right and increases from top to bottom
(D) Increases from left to right and decreases from top to bottom

CPT163

Types of Oxides

46. Which of the following does not act as an amphoteric oxide?
(A) Al_2O_3 (B) ZnO (C) BeO (D) NO

CPT164

For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

Atomic Radius ; Trends for d- and f- Block Elements

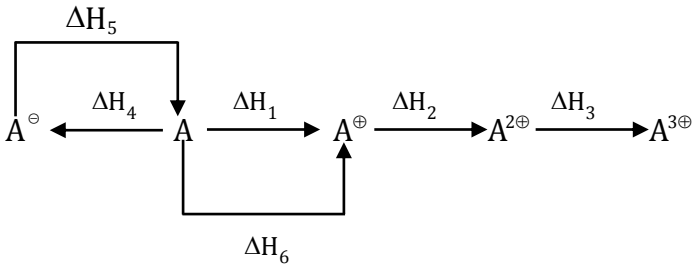
1. Select correct order of size :

(A) $Ti^{2\oplus} < Ti < Zr$ (B) $Ti^{2\oplus} < Ti < Hf$ (C) $Zr^{2\oplus} < Zr \approx Hf$ (D) $Hf^{2\oplus} < Hf \approx Zr$

CPT184

Definition of Ionization Energy; Successive Ionization Energy

2. Consider the following sequence of reaction



If electronic configuration of element A is $[Ne]3s^1$, then which of the following order are correct regarding given enthalpies ?

(A) $|\Delta H_4| = |\Delta H_5|$ (B) $|\Delta H_2| > |\Delta H_1|$

(C) $|\Delta H_2| > |\Delta H_3|$ (D) $|\Delta H_1| = |\Delta H_6|$

Rankers Academy JEE

CPT142

Periodicity of Electron Affinity in s and p block elements

3. Which of the following are correct?

(A) IE_2 (Mg) < IE_2 (Na) (B) EA (N) < EA (P)

(C) Atomic size Mg^{+2} > Atomic size ($Li^{\oplus}$) (D) IP of Na < Mg < Al

CPT186

2nd Electron Affinity

4. Which of the following is/are correct?

(A) For $A(g) + e^{\ominus} \longrightarrow A^{\ominus}(g)$ ΔH may be negative

(B) For $A^{\ominus}(g) + e^{\ominus} \longrightarrow A^{2\ominus}(g)$ ΔH may be negative

(C) For $A^{\ominus}(g) + e^{\ominus} \longrightarrow A^{2\ominus}(g)$ ΔH must be positive

(D) For $Ne(g) + e^{\ominus} \longrightarrow Ne^{\ominus}(g)$ ΔH may be zero

CPT143

Different Scales to Measure Electronegativity

5. An element 'E' have IE = x eV/atom and EA = y eV / atom and EN on Pauling scale is 1.2. Find EN of 'E' on Mulliken scale :

(A) $\frac{x+y}{2}$ (B) 1.2×2.8 (C) $\frac{x+y}{2.8} - 1.2$ (D) $1.2 - \frac{x+y}{2}$

CPT156

For More Join: @IITJEE_Advanced

Applications of Electronegativity; Acidic Strength of Oxides and Hydroxides

6. Which of the following order(s) is / are **CORRECT**:

(A) Li < Be < B < C (IE_1)

(B) HF < HCl < HBr < HI (Bond length)

(C) $Na_2O < MgO < Al_2O_3 < SiO_2 < P_2O_5$ (Acidic)

(D) $Li^{\oplus}$ (g) < $Na^{\oplus}$ (g) < $K^{\oplus}$ (g) < $Cs^{\oplus}$ (g) (Ionic radius)

CPT187

COMPREHENSION TYPE QUESTIONS

Factors Affecting Electronegativity; Periodicity of Electronegativity in s-p- block elements

Paragraph for Question 7 to 8

Four elements P, Q, R & S have ground state electronic configuration as:

P $\rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$ Q $\rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$

R $\rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^3$ S $\rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^1$

7. Comment which of the following option represent the correct order of true (T) & false (F) statement.

I. size of P < size of Q

II. size of R < size of S

III. size of P < size of R (appreciable difference)

IV. size of Q < size of S (appreciable difference)

(A) TTTT (B) TTTF (C) FFTT (D) TTFF

CPT188

Rankers Academy JEE

8. Order of IE_1 values among the following is

(A) P > R > S > Q (B) P < R < S < Q (C) R > S > P > Q (D) P > S > R > Q

CPT189

Applications of Electronegativity; Polarity of Bond; Calculation of Percentage Ionic Character

Paragraph for Question No. 9 to 10

Let covalent length of A–B molecule is denoted by d_{A-B} and r_A and r_B be the covalent radius of A and B respectively.

9. If d_{A-A} = 2 Å, d_{B-B} = 1 Å and d_{A-B} = 1.23 Å then the difference in electronegativities of A & B in Pauling Scale is

(A) 3 (B) 8.4 (C) 5.6 (D) 2.8

CPT157

10. The percentage covalent character for the data given in above question is

(Assume EN in Pauling scale)

(A) 58.5 (B) 79.5 (C) 20.5 (D) 48

CPT158

Nature of Hydroxides in Aqueous Medium

Paragraph for Question 11 and 12

Nature of bond can be predicted on the basis of electronegativity of bonded atoms, greater difference in electronegativity (X), more will be the polarity of bond, and polar bond are easily broken in polar solvent like water. For hydroxy acids X_O–X_A difference predict the nature of oxide formed by the element A.

For More Join: @IITJEE_Advanced

$|X_O - X_A| > |X_O - X_H|$ then A–O–H show basic nature (NaOH)

$|X_O - X_A| < |X_O - X_H|$ then A–O–H show acidic nature (H–O–Cl)

With the help of EN values [EN_A = 1.8, EN_B = 2.6, EN_C = 1.6, EN_D = 2.8] answer the following questions for the compounds AOH, BOH, COH, DOH.

11. Compounds whose aqueous solution is acidic and order of their acidic strength

(A) AOH, COH ; AOH < COH (B) DOH, BOH; DOH > BOH

(C) AOH, COH ; AOH > COH (D) DOH, BOH; DOH < BOH

CPT173

12. Compounds whose aqueous solution is basic and order of their basic strength

(A) AOH, COH ; AOH < COH (B) DOH, BOH ; DOH > BOH

(C) AOH, COH ; AOH > COH (D) DOH, BOH ; DOH < BOH

CPT174

MATRIX MATCH TYPE QUESTION

Electronic Configuration of Elements and the periodic table, Electronic Configuration in Period; Groupwise Electronic Configuration, Identification of group, block and period of an element

13.

Column-I	**Column-II**
Element	**Period and group number respectively**
(P) Si	(1) 4, 3
(Q) Sc	(2) 3, 14
(R) Ga	(3) 6, 13
(S) Tℓ	(4) 4, 13

Rankers Academy JEE

CPT190

Code :

	P	**Q**	**R**	**S**
(A)	2	1	3	4
(B)	1	2	4	3
(C)	2	1	4	3
(D)	4	3	1	2

Factors Affecting Electronegativity; Periodicity of Electronegativity in s-p- block elements

14. Match the column :

Column I	**Column II**
(A) Cl	(P) Metal
(B) F	(Q) Highest negative electron gain enthalpy
(C) Cu	(R) Most Electronegative element
(D) He	(S) Highest ionisation energy.

CPT191

For More Join: @IITJEE_Advanced

EXERCISE (S)

Part 1 : JEE Main (Single Integer type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

Neil Bohr's classification; Some commonly used terms

1. Find the number of transition elements in the following.
Zn, Cd, Hg, Pt, U, Sn

CPT198

Nomenclature of Elements with Atomic Number >100; Notation for IUPAC Nomenclature of Elements

2. Find out the sum of all the digits of atomic number of the elements Uuq(Fl).
If your answer is greater than 9 then add all the digits.
For example. Ans. 12 then fill 1 + 2 = 3, Ans. 123 then fill 1 + 2 + 3 = 6 and Ans. 20 then fill 2 + 0 = 2

CPT197

Electronic Configuration of Elements and the periodic table, Electronic Configuration in Period; Groupwise Electronic Configuration, Identification of group, block and period of an element

3. Find the sum of total number of 5f-electrons in Th and Ac.

CPT199

Calculation of Shielding Constant using Slater's Rule

4. Maximum Number of electrons in sulphur which has same value of Z_{eff}

CPT108

Rankers Academy JEE

5. Find the total number of electrons in Cl which have Z_{eff} less than 10, when σ is calculated by Slater rules.

CPT192

Ionic Radius; Factors Affecting Ionic Radius

6. Among the following species, how many have their ionic size greater than $S^{2\ominus}$
$O^{2\ominus}$, $P^{3\ominus}$, $Cl^{1\ominus}$, $S^{1\ominus}$, $Ca^{2\oplus}$, $K^{\oplus}$?

CPT202

7. How many of the following radius orders are incorrect?

(i) $N^{3\ominus} > P^{3\ominus}$ (ii) $O^{2\ominus} > F^{\ominus}$ (iii) $Ca^{2\oplus} > Sr^{2\oplus}$

(iv) $S^{1\ominus} > S^{2\ominus}$ (v) $S^{2\ominus} > O^{\ominus}$ (vi) $Na^{\oplus} > F^{\ominus}$

CPT026

Definition of Ionization Energy; Successive Ionization Energy

8. Consider ionization of an element "E" :

$$E_{(g)} \xrightarrow{+1P_1} E^{\oplus}_{(g)} \xrightarrow{+1P_2} E^{2\oplus}_{(g)} \xrightarrow{+1P_3} E^{3\oplus}_{(g)}, \xrightarrow{+1P_4} E^{4\oplus}_{(g)} \xrightarrow{+1P_5} E^{*}_{(g)}$$

If $E^{5\oplus}$ does not have any electron. Find number of electrons in valence shell of element "E".

CPT144

Periodicity of Ionization Energy in s and p Block Elements

9. Total number of elements which have less IE_1 than that of 'N'.
Be, B, C, F, P, He

CPT125

10. Upto argon find the number of elements which have lower IE_1 as compared to He.

CPT193

For More Join: @IITJEE_Advanced

11. How many of the following species have lower IE_1 value compared to $F^{\ominus}$?
$Cl^{\ominus}$, $Br^{\ominus}$, $O^{\ominus}$, $I^{\ominus}$

CPT194

12. Upto argon find the number of elements which have lower IE_1 as compared to He mark your answer as sum of digits

CPT023

Periodicity of Electron Affinity in s and p block elements

13. The number of pairs, in which EA of the second element is more than that of the first element is :
[O, S], [C, N], [O, N], [N, P], [$Cl^{\oplus}$, $F^{\oplus}$], [$K^{\oplus}$, $Na^{\oplus}$]

CPT195

Relation between Ionization Energy and Electron Affinity

14. $A^{\oplus}_{(g)} \rightarrow A(g)\ \Delta H = -2ev$

$2A(g) \rightarrow 2A^{\oplus}(g)\ \Delta H = x$
Find the value of x in eV.

CPT196

Factors Affecting Electronegativity; Periodicity of Electronegativity in s-p- block elements

15. Choose the number of CORRECT statement(s) from the following
(a) 1st ionization potential of 'B' is higher than that of 'Be'.
(b) Electron affinity of 'O' is higher than that of 'S'.
(c) [Ar] $4s^2\ 3d^3$ is the electronic configuration of $Mn^{2\oplus}$.
(d) 2nd ionization potential of Na > 1st ionization potential of Na.
(e) 1st ionization potential of N > 2nd ionization potential of Na.
(f) Electronegativity of Cl > Electronegativity of F.
(g) $C \longrightarrow C^{2\oplus}$ change is called 2nd ionisation potential of carbon.
(h) Energy is required to convert $He \longrightarrow He^{\ominus}$
(i) Conversion of $O \longrightarrow O^{2\ominus}$ is exothermic.

Bansal's Academy JEE

CPT200

16. Count the number of properties, which have higher values for 'O' atom than that of 'S'.
(a) Ionisation potential
(b) Electron affinity
(c) Covalent radius
(d) Electronegativity
(e) (Proton : Electron) ratio

CPT201

Different Scales to Measure Electronegativity

17. I.P. of an element is equal to 14.2 eV and EA is equal to 2.6 eV, then find electronegativity according to Pauling scale.

CPT159

Applications of Electronegativity; Acidic Strength of Oxides and Hydroxides

18. The number of oxygen atoms in the most acidic oxide among Cl_2O, Cl_2O_3, ClO_2, Cl_2O_5, Cl_2O_6, Cl_2O_7 is.

CPT027

Types of Oxides

19. How many of the following oxides are amphoteric
BeO, Al_2O_3, SnO_2, N_2O, NO, CO_2, PbO_2, Sb_2O_3

CPT028

For More Join: @IITJEE_Advanced

Part 2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

Long Form of Periodic Table

20. If there are 9 periods in a periodic table and each orbital can contain maximum 5 electrons, then how many maximum numbers of elements will be present in period 9?
[Write your answer by dividing it with 100]

CPT203

21. Maximum number of elements that can be accommodated in 10th period of periodic table, if it is possible is :

CPT204

22. The atomic number of elements "Unnilquadium" is X. The value of $\left(\frac{X-4}{30}\right)$ is-

CPT205

Calculation of Shielding Constant using Slater's Rule

23. According to Slater's rule, calculate effective nuclear charge (Z_{eff}) on last electron of F^-.

CPT206

Relation between Ionization Energy and Electron Affinity

24. Given : $Na \longrightarrow Na^+ + e^-$; $\Delta H = \frac{5eV}{atom} = 482.5\frac{KJ}{mole}$
Find Electronegativity of Cl on Mullikan scale if
E.A. ($C\ell$) = $347.4\frac{KJ}{mole}$ and E.A. (Cl^+) = $1254.5\frac{KJ}{mole}$

Rankers Academy JEE

CPT207

Different Scales to Measure Electronegativity

25. Given :
Bond energy of following bonds :
F – F = 38 kcal/mol
Cl – Cl = 58 kcal/mol
Cl – F = 61 kcal/mol
Electronegativity of F = 4
Calculate electronegativity of Cl atom.

CPT208

26. The EN of an atom 'y' on Pauling scale is 3.06. Then EN of y on Alfred Rochow's scale is ___

CPT209

27. Electronegativity of an element is 1 on the Pauling scale, then it's value on Mulliken scale :

CPT210

Applications of Electronegativity; Polarity of Bond; Calculation of Percentage Ionic Character

28. If bond length of F_2 = 1.44Å, Bond length of H_2 = 0.74Å. Find out the bond length of H – F ? (EN of F is 4.0, EN of H is 2.1)

CPT211

29. % Ionic character in the Covalent bond A – B is - (Given that X_A = 2, X_B = 3)

CPT212

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. Given below are two statements :

Statement (I) : According to the Law of Octaves, the elements were arranged in the increasing order of their atomic number.

Statement (II) : Meyer observed a periodically repeated pattern upon plotting physical properties of certain elements against their respective atomic numbers.

In the light of the above statements, choose the **correct** answer from the options given below :

[JEE (Main) 2025]

(1) **Statement I** is false but **Statement II** is true
(2) Both **Statement I** and **Statement II** are true
(3) **Statement I** is true but **Statement II** is false
(4) Both **Statement I** and **Statement II** are false

CPT228

2. Given below are two statements about X-ray spectra of elements :

Statement (I) : A plot of $\sqrt{\nu}$ (ν = frequency of X-rays emitted) vs atomic mass is a straight line.

Statement (II) : A plot of ν (ν = frequency of X-rays emitted) vs atomic number is a straight line.

In the light of the above statements choose the **correct** answer from the options given below :

[JEE (Main) 2025]

(1) **Statement I** is true but **Statement II** is false
(2) Both **Statement I** and **Statement II** are true
(3) Both **Statement I** and **Statement II** are false
(4) **Statement I** is false but **Statement II** is true

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CPT229

3. Which of the following Statements are NOT true about the periodic table?

A. The properties of elements are function of atomic weights.
B. The properties of elements are function of atomic numbers.
C. Elements having similar outer electronic configuration are arranged in same period.
D. An element's location reflects the quantum numbers of the last filled orbital.
E. The number of elements in a period is same as the number of atomic orbitals available in energy level that is being filled.

Choose the correct answer from the options given below: **[JEE (Main) 2025]**

(1) A, C and E Only (2) D and E Only (3) A and E Only (4) B, C and E Only

CPT230

4. The metal that has very low melting point and its periodic position is closer to a metalloid is :

[JEE (Main) 2022]

(1) Al (2) Ga (3) Se (4) In

CPT049

5. If IUPAC name of an element is "Unununium" then the element belongs to nth group of periodic table. The value of n is_____ **[JEE (Main) 2024]**

CPT213

For More Join: @IITJEE_Advanced

6. The IUPAC symbol for the element with atomic number 119 would be : **[JEE (Main) 2019]**

(1) Unh (2) Uun (3) Une (4) Uue

CPT033

7. Given below are two statements:

Statement (I) : For a given shell, the total number of allowed orbitals is given by n^2.

Statement (II) : For any subshell, the spatial orientation of the orbitals is given by $-l$ to $+l$ values including zero.

In the light of the above statements, choose the **correct** answer from the options given below: **[JEE (Main) 2025]**

(1) **Statement I** is true but **Statement II** is false

(2) **Statement I** is false but **Statement II** is true

(3) Both **Statement I** and **Statement II** are true

(4) Both **Statement I** and **Statement II** are false

CPT231

8. The element that does not belong to the same period of the remaining elements (modern periodic table) is: **[JEE (Main) 2025]**

(1) Palladium (2) Iridium (3) Osmium (4) Platinum

CPT232

9. The incorrect decreasing order of atomic radii is : **[JEE (Main) 2025]**

(1) Mg > Al > C > O (2) Al > B > N > F (3) Be > Mg > Al > Si (4) Si > P > Cl > F

CPT233

10. The increasing order of the atomic radii of the following elements is : **[JEE (Main) 2020]**

(a) C (b) O (c) F (d) Cl (e) Br

(1) (b) < (c) < (d) < (a) < (e) (2) (a) < (b) < (c) < (d) < (e)

(3) (d) < (c) < (b) < (a) < (e) (4) (c) < (b) < (a) < (d) < (e)

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CPT047

11. The correct order of the atomic radii of C, Cs, Al and S is : **[JEE (Main) 2019]**

(1) S < C < Al < Cs (2) S < C < Cs < Al

(3) C < S < Cs < Al (4) C < S < Al < Cs

CPT031

12. The statement(s) that are **correct** about the species O^{2-}, F^-, Na^+ and Mg^{2+}.

(A) All are isoelectronic

(B) All have the same nuclear charge

(C) O^{2-} has the largest ionic radii

(D) Mg^{2+} has the smallest ionic radii

Choose the **most appropriate** answer from the options given below : **[JEE (Main) 2024]**

(1) (B), (C) and (D) only (2) (A), (B), (C) and (D)

(3) (C) and (D) only (4) (A), (C) and (D) only

CPT214

13. The correct increasing order of the ionic radii is **[JEE (Main) 2023]**

(1) $Cl^- < Ca^{2+} < K^+ < S^{2-}$ (2) $K^+ < S^{2-} < Ca^{2+} < Cl^-$

(3) $S^{2-} < Cl^- < Ca^{2+} < K^+$ (4) $Ca^{2+} < K^+ < Cl^- < S^{2-}$

CPT178

For More Join: @IITJEE_Advanced

14. The isoelectronic set of ions is : **[JEE (Main) 2019]**

(1) $N^{3\ominus}, Li^{\oplus}, Mg^{2\oplus}$ and $O^{2\ominus}$ (2) $Li^{\oplus}, Na^{\oplus}, O^{2\ominus}$ and $F^{\ominus}$

(3) $F^{\ominus}, Li^{\oplus}, Na^{\oplus}$ and $Mg^{2\oplus}$ (4) $N^{3\ominus}, O^{2\ominus}, F^{\ominus}$ and $Na^{\oplus}$

CPT034

15. Given below are two statements :

Statement (I) : The first ionization energy of Pb is greater than that of Sn

Statement (II) : The first ionization energy of Ge is greater than that of Si.

In the light of the above statements, choose the **correct** answer from the options given below : **[JEE (Main) 2025]**

(1) **Statement I** is true but **Statement II** is false

(2) Both **Statement I** and **Statement II** are false

(3) **Statement I** is false but **Statement II** is true

(4) Both **Statement I** and **Statement II** are true

CPT234

16. The successive 5 ionisation energies of an element are 800, 2427, 3658, 25024 and 32824 kJ/mol, respectively. By using the above values predict the group in which the above element is present : **[JEE (Main) 2025]**

(1) Group 2 (2) Group 13 (3) Group 4 (4) Group 14

CPT235

17. The element having the highest first ionization enthalpy is **[JEE (Main) 2024]**

(1) Si (2) Al (3) N (4) C

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CPT215

18. The first ionization energy (in kJ/mol) of Na, Mg, Al and Si respectively, are : **[JEE (Main) 2020]**

(1) 496, 737, 577, 786 (2) 786, 737, 577, 496

(3) 496, 577, 737, 786 (4) 496, 577, 786, 737

CPT037

19. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R.**

Assertion A: The energy required to form Mg^{2+} from Mg is much higher than that required to produce Mg^{+}.

Reason R: Mg^{2+} is small ion and carry more charge than Mg^{+}.

In the light of the above statements, choose the **correct** answer from the options given below: **[JEE (Main) 2023]**

(1) Both A and R are true but R is NOT the correct explanation of A.

(2) A is true but R is false.

(3) A is false but R is true.

(4) Both A and R are true and R is the correct explanation of A.

CPT176

20. The correct decreasing order for metallic character is **[JEE (Main) 2022]**

(1) Na > Mg > Be > Si > P (2) P > Si > Be > Mg > Na

(3) Si > P > Be > Na > Mg (4) Be > Na > Mg > Si > P

CPT050

For More Join: @IITJEE_Advanced

21. Given below are two statements: **[JEE (Main) 2024]**

Statement I: Fluorine has most negative electron gain enthalpy in its group.

Statement II: Oxygen has least negative electron gain enthalpy in its group.

In the light of the above statements, choose the most appropriate from the options given below.

(1) Both Statement I and Statement II are true

(2) Statement I is true but Statement II is false

(3) Both Statement I and Statement II are false

(4) Statement I is false but Statement II is true

CPT216

22. Inert gases have positive electron gain enthalpy. Its correct order is **[JEE (Main) 2023]**

(1) Xe < Kr < Ne < He (2) He < Ne < Kr < Xe

(3) He < Xe < Kr < Ne (4) He < Kr < Xe < Ne

CPT177

23. The electron gain enthalpy (in kJ/mol) of fluorine, chlorine, bromine and iodine, respectively are : **[JEE (Main) 2020]**

(1) – 333, – 349, – 325 and – 296 (2) –296, – 325, – 333 and – 349

(3) – 333, – 325, – 349 and – 296 (4) –349, – 333, – 325 and – 296

CPT040

24. Within each pair of elements of F & Cl, S & Se, and Li & Na, respectively, the elements that release more energy upon an electron gain are- **[JEE (Main) 2020]**

(1) F, Se and Na (2) F, S and Li (3) Cl, S and Li (4) Cl, Se and Na

CPT041

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25. An element 'E' has the ionisation enthalpy value of 374 kJ mol^{-1}. 'E' reacts with elements A, B, C and D with electron gain enthalpy values of –328, –349, –325 and –295 kJ mol^{-1}, respectively.

The correct order of the products EA, EB, EC and ED in terms of ionic character is : **[JEE (Main) 2025]**

(1) EB > EA > EC > ED (2) ED > EC > EA > EB (3) EA > EB > EC > ED (4) ED > EC > EB > EA

CPT236

26. Given below are two statements :

Statement (I) : The radii of isoelectronic species increases in the order.

$Mg^{2+} < Na^{+} < F^{-} < O^{2-}$

Statement (II) : The magnitude of electron gain enthalpy of halogen decreases in the order.

Cl > F > Br > I

In the light of the above statements, choose the **most appropriate answer** from the options given below : **[JEE (Main) 2025]**

(1) **Statement I** is incorrect but **Statement II** is correct

(2) Both **Statement I** and **Statement II** are incorrect

(3) **Statement I** is correct but **Statement II** is incorrect

(4) Both **Statement I** and **Statement II** are correct

CPT237

27. Which of the following electronegativity order is **incorrect**? **[JEE (Main) 2025]**

(1) Al < Mg < B < N (2) Al < Si < C < N (3) Mg < Be < B < N (4) S < Cl < O < F

CPT238

For More Join: @IITJEE_Advanced

28. Match **List-I** with **List-II**

List-I		List-II	
(A)	$Al^{3+} < Mg^{2+} < Na^{+} < F^{-}$	(I)	Ionisation Enthalpy
(B)	B < C < O < N	(II)	Metallic character
(C)	B < Al < Mg < K	(III)	Electronegativity
(D)	Si < P < S < Cl	(IV)	Ionic radii

Choose the **correct** answer from the options given below : **[JEE (Main) 2025]**

(1) A-IV, B-I, C-III, D-II (2) A-II, B-III, C-IV, D-I

(3) A-IV, B-I, C-II, D-III (4) A-III, B-IV, C-II, D-I

CPT239

29. Match **List I** with **List II**

List-I (Elements)		List-II (Properties in their respective groups)	
A.	Cl, S	I.	Elements with highest electronegativity
B.	Ge, As	II.	Elements with largest atomic size
C.	Fr, Ra	III	Elements which show properties of both metals and non metal
D.	F, O	IV	Elements with highest negative electron gain enthalpy

Choose the **correct** answer from the options given below : **[JEE (Main) 2024]**

(1) A-II, B-III, C-IV, D-I (2) A-III, B-II, C-I, D-IV

(3) A-IV, B-III, C-II, D-I (4) A-II, B-I, C-IV, D-III

CPT217

30. In general, the property (magnitudes only) that shows an opposite trend in comparison to other properties across a period is **[JEE (Main) 2020]**

(1) Electronegativity (2) Electron gain enthalpy

(3) Ionization enthalpy (4) Atomic radius

CPT042

31. The correct option with respect to the Pauling electronegativity values of the elements is :- **[JEE (Main) 2019]**

(1) Ga < Ge (2) Si < Al (3) P > S (4) Te > Se

CPT032

32. The electronegativity of aluminium is similar to : **[JEE (Main) 2019]**

(1) Boron (2) Carbon

(3) Lithium (4) Beryllium

CPT036

33. Among the statements (I – IV), the correct ones are :

(I) Be has smaller atomic radius compared to Mg.

(II) Be has higher ionization enthalpy than Al.

(III) Charge/radius ratio of Be is greater than that of Al.

(IV) Both Be and Al form mainly covalent compounds. **[JEE (Main) 2020]**

(1) (I), (II) and (IV) (2) (II), (III) and (IV)

(3) (I), (II) and (III) (4) (I), (III) and (IV)

CPT044

For More Join: @IITJEE_Advanced

34. Given below are two statements :

Statement (I) : An element in the extreme left of the periodic table forms acidic oxides.

Statement (II) : Acid is formed during the reaction between water and oxide of a reactive element present in the extreme right of the periodic table.

In the light of the above statements, choose the **correct** answer from the options given below :

[JEE (Main) 2025]

(1) **Statement-I** is false but **Statement-II** is true.

(2) Both **Statement-I** and **Statement-II** are false.

(3) **Statement-I** is true but **Statement-II** is false.

(4) Both **Statement-I** and **Statement-II** are true.

CPT240

35. The acidic, basic and amphoteric oxides, respectively, are : **[JEE (Main) 2020]**

(1) MgO, Cl_2O, Al_2O_3

(2) Cl_2O, CaO, P_4O_{10}

(3) Na_2O, SO_3, Al_2O_3

(4) N_2O_3, Li_2O, Al_2O_3

CPT038

36. Three elements X, Y and Z are in the 3rd period of the periodic table. The oxides of X, Y and Z, respectively, are basic, amphoteric and acidic. The correct order of the atomic numbers of X, Y and Z is : **[JEE (Main) 2020]**

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(1) Z < Y < X

(2) X < Z < Y

(3) X < Y < Z

(4) Y < X < Z

CPT043

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Advanced) PYQ

1. The increasing order of atomic radii of the following group 13 elements is **[JEE (Advanced) 2016]**

(A) Al < Ga < In < Tl (B) Ga < Al < In < Tl (C) Al < In < Ga < Tl (D) Al < Ga < Tl < In

CPT076

2. The correct order of radii is : **[JEE (Advanced) 2000]**

(A) N < Be < B (B) $F^{\ominus} < O^{2\ominus} < N^{3\ominus}$

(C) Na < Li < K (D) $Fe^{3\oplus} < Fe^{2\oplus} < Fe^{4\oplus}$

CPT072

3. The IE_1 of Be is greater than that of B. [T/F] **[JEE (Advanced) 2001]**

CPT073

4. The set representing correct order of IP_1 is **[JEE (Advanced) 2001]**

(A) K > Na > Li (B) Be > Mg > Ca (C) B > C > N (D) Fe > Si > C

CPT074

5. The 1st, 2nd and the 3rd ionization enthalpies I_1, I_2 and I_3, of four atoms with atomic numbers n, n+1, n+2 and n+3, where n < 10, are tabulated below. What is the value of n?

[JEE (Advanced) 2020]

Atomic number	Ionization Enthalpy (kJ/mol)		
	I_1	I_2	I_3
n	1681	3374	6050
n+1	2081	3952	6122
n+2	496	4562	6910
n+3	738	1451	7733

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CPT078

6. **Statement-1 :** F atom has a less negative electron gain enthalpy than Cl atom.

Statement-2 : Additional electron is repelled more efficiently by 3p electron in Cl atom than by 2p electron in F atom. **[JEE (Advanced) 2000]**

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

CPT071

7. Identify the least stable ion amongst the following : **[JEE (Advanced) 2002]**

(A) $Li^{\ominus}$ (B) $Be^{\ominus}$ (C) $B^{\ominus}$ (D) $C^{\ominus}$

CPT075

8. The option(s) with only amphoteric oxides is (are): **[JEE (Advanced) 2017]**

(A) Cr_2O_3, CrO, SnO, PbO (B) NO, B_2O_3, PbO, SnO_2

(C) Cr_2O_3, BeO, SnO, SnO_2 (D) ZnO, Al_2O_3, PbO, PbO_2

CPT077

For More Join: @IITJEE_Advanced

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	C	A	C	C	C	D	A	A	D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	C	B	D	C	C	B	A	C	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	A	B	B	A	A	B	D	C	A
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	D	B	B	C	C	C	B	A	B	A
Que.	41	42	43	44	45	46				
Ans.	C	B	C	C	C	D				

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	ABCD	ABD	AB	AC	AB	BCD	B	A	A	C
Que.	11	12	13	14						
Ans.	B	A	C	[(A) Q ; (B) R ; (C) P ; (D) S]						

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EXERCISE (S)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	6	0	8	7	1	4	3	4	17
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	8	4	4	2	2	3	7	5	1.25
Que.	21	22	23	24	25	26	27	28	29	
Ans.	72.00	3.33	4.85	8.30	3.20-3.23	2.32	2.80	0.90-0.93	19.50	

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	3	1	2	11	4	3	1	3	4
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	4	4	4	1	2	3	1	4	1
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	4	3	1	3	1	4	1	3	3	4
Que.	31	32	33	34	35	36				
Ans.	1	4	1	1	4	3				

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	
Ans.	B	B	T	B	9	C	B	CD	

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